

David Jiang

 david-jiang.com  david.jiang1@uwaterloo.ca  github.com/DVADOSO  linkedin.com/in/davidjiangdev

EDUCATION

University of Waterloo | *Honours Computer Science (co-op)*

GPA: 3.96

Awards: Canadian Computing Competition Senior Division 95th percentile, Euclid Math Contest Distinction

SKILLS

Languages: TypeScript, Python, JavaScript, C#, C++, C, SQL, Ruby

Frameworks & APIs: React, Next.js, Node.js, .NET Core, FastAPI, Ruby on Rails, Redux, RTK Query

Tools & Cloud: Git, Docker, Azure (Web Apps, Pipelines, AD), PostgreSQL, MS Server, Linux, Bash, OpenCV

EXPERIENCE

Full Stack Software Developer Intern

Jan. 2026 – Apr. 2026

API Group Inc. (Habtech)

Mississauga, ON

- Designed and built an **end-to-end quotation system** across **Next.js** and **.NET Core** (quote creation, pricing, approvals, document upload), replacing a third-party platform and **saving ~\$2M** in licensing costs
- Architected a **configurable workflow engine** with **role-based state transitions** and **step-level permission enforcement**, enabling non-technical users to define multi-step workflows without code changes
- Implemented a **multi-tenant RBAC system**, supporting tenant-scoped user management, **permission hierarchies**, and endpoint-level **authorization guards**, enabling secure access control on a shared platform
- Engineered a **smart-diff algorithm** for bulk imports that compares incoming records against existing database to issue only necessary **CRUD** operations, eliminating redundant writes and **avoiding an Azure tier upgrade**
- Containerized platform services with **Docker** deployed to **Azure Web Apps** via **CI/CD pipelines** for consistent staging-to-production releases with coordinated **database migrations**

Software Engineering Intern

May 2025 – Aug. 2025

StackAdapt

Toronto, ON

- Replaced an underperforming third-party ad-vetting service with an **auto-approval system** built on standardized ad categorization and **feature-flagged rollout**, restoring the product team's ability to consistently meet SLA
- Modernized a **legacy Ruby on Rails internal tool** by rebuilding the frontend in **React**, replacing raw HTML views with a component-driven UI to significantly improve daily experience for product reviewers
- Designed a **RTK data layer** for ad campaign creation, building **React hooks**, **RTK Query endpoints**, and **pagination/sorting logic** to support campaign configuration including budgeting and organization
- Reclaimed **90%+ of database storage** by automating cleanup of stale records older than 30 days across a database of **100k+ entries**, eliminating storage pressure and improving query latency
- Cut **Sentry error volume by 40%+** by filtering noise from genuine errors and adding **retry-with-delay logic** for transient post-deletion DB sync failures, reducing on-call burden across the engineering team

PROJECTS

Camera Obstruction & Movement Detection App | *FastAPI, Next.js, PostgreSQL, OpenCV*

[Source code](#)

- Built a **full-stack CV dashboard** that flags camera tampering by comparing live frames against reference images using **SSIM**, **ORB feature matching**, and **RANSAC homography inlier counts** as similarity signals
- Designed an **auto-tuning service** that searches detection thresholds against user labels to **maximize F1**, achieving **94.3% accuracy** on 500 labeled samples, replacing manually picked thresholds
- Engineered a **synthetic data generator** that applies transformation types (occlusion, rotation, perspective, noise) to given base image, letting users tune detection thresholds without first collecting dozens of real captures
- Containerized **FastAPI**, **Next.js**, and **PostgreSQL** services via **Docker Compose** with **Alembic migrations** for persistent image storage and hot-reload development

Full Stack Task Planning Web Application | *SvelteKit, Google Cloud Platform*

[Source code](#)

- Built a **full-stack task management web app**, organizing user tasks on an **Eisenhower matrix** (importance vs. urgency) with hover-to-reveal descriptions, deployed to Firebase Hosting
- Implemented **Firebase Auth and Realtime Database** for persistent multi-device task sync, alongside a **guest mode** backed by in-memory **Svelte stores** for friction-free trial access without account creation